

Bài 3: Code

```
Explorer
  SOA
    api_session_02
    api_session_01
    .venv
      Include
      Lib
      Scripts
      pyvenv.cfg
    app.py
    ex_Week1
    ex_Week2
    SOA - B1_ Gi...
    SOA - B2_ Ki...

requirements.txt
app.py
list_books

1 from flask import Flask, jsonify, request, make_response
2 app = Flask(__name__)
3 BOOKS = []
4 _next_id = 1
5 # Tham số phân trang mặc định
6 DEFAULT_SIZE, MAX_SIZE = 20, 100
7 @app.get("/books")
8 def list_books():
9     try:
10         page = int(request.args.get("page", 1))
11         size = int(request.args.get("size", DEFAULT_SIZE))
12     except ValueError:
13         return jsonify(error="page and size must be int"), 400
14
15 size = max(min(size, MAX_SIZE), 1)
16 # Filter: author chính xác, q tìm kiếm trong title
17 flt = BOOKS
18 a = request.args.get("author")
19 if a:
20     flt = [b for b in flt if b["author"].lower() == a.lower()]
21 q = (request.args.get("q") or "").lower()
22 if q:
23     flt = [b for b in flt if q in b["title"].lower()]
24
25 # Paginate
26 total = len(flt)
27 start = (page - 1) * size
28 end = start + size
29 items = flt[start:end]
30 last = (total + size - 1) // size
31 # HATEOAS links
32 def u(p):
33     return f"/books?page={p}&size={size}"
34 links = {
35     "self": {"href": u(page)},
36     "first": {"href": u(1)},
37     "last": {"href": u(max(last, 1))}
38 }
39 if page > 1:
40     links["prev"] = {"href": u(page - 1)}
41 if end < total:
42     links["next"] = {"href": u(page + 1)}
43 body = {
44     "data": items,
45     "pagination": {
46         "page": page,
47         "size": size,
48         "total": total,
49         "total_pages": last
50     },
51     "_links": links
52 }
53 resp = make_response(jsonify(body), 200)
54 resp.headers["Cache-Control"] = "public, max-age=30"
55 return resp
56
57 if __name__ == "__main__":
58     app.run(debug=True)
```

1. Test Panigation(Phân trang)

```
(root)  
THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)  
● $ curl -i "localhost:5000/books?page=1&size=10"  
HTTP/1.1 200 OK  
Server: Werkzeug/3.1.8 Python/3.12.3  
Date: Fri, 18 Sep 2026 15:12:24 GMT  
Content-Type: application/json  
Content-Length: 304  
Cache-Control: public, max-age=30  
Connection: close  
  
{  
  "_links": {  
    "first": {  
      "href": "/books?page=1&size=10"  
    },  
    "last": {  
      "href": "/books?page=1&size=10"  
    },  
    "next": {  
      "href": "/books?page=1&size=10"  
    }  
  },  
  "data": [],  
  "pagination": {  
    "page": 1,  
    "size": 10,  
    "total": 0,  
    "total_pages": 0  
  }  
}
```

2. Test filtering by author

```

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)
● $ curl -i "localhost:5000/books?author=R.+Martin"
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 15:13:37 GMT
Content-Type: application/json
Content-Length: 304
Cache-Control: public, max-age=30
Connection: close

{
  "_links": {
    "first": {
      "href": "/books?page=1&size=20"
    },
    "last": {
      "href": "/books?page=1&size=20"
    },
    "self": {
      "href": "/books?page=1&size=20"
    }
  },
  "data": [],
  "pagination": {
    "page": 1,
    "size": 20,
    "total": 0,
    "total_pages": 0
  }
}

```

3. Test Query search(tìm kiếm theo tên sách)

```

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)
● $ curl -i "localhost:5000/books?q=clean"
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 15:15:05 GMT
Content-Type: application/json
Content-Length: 304
Cache-Control: public, max-age=30
Connection: close

```

```
{
  "_links": {
    "first": {
      "href": "/books?page=1&size=20"
    },
    "last": {
      "href": "/books?page=1&size=20"
    },
    "self": {
      "href": "/books?page=1&size=20"
    }
  },
  "data": [],
  "pagination": {
    "page": 1,
    "size": 20,
    "total": 0,
    "total_pages": 0
  }
}
```

4. Test Header Accept

```
THI@LAPTOP-HCT3QG47 MINGW64 /e/soa (main)
● $ curl -H "Accept: application/json" localhost:5000/books
{
  "_links": {
    "first": {
      "href": "/books?page=1&size=20"
    },
    "last": {
      "href": "/books?page=1&size=20"
    },
    "self": {
      "href": "/books?page=1&size=20"
    }
  },
  "data": [],
  "pagination": {
    "page": 1,
    "size": 20,
    "total": 0,
    "total_pages": 0
  }
}
```